

Behavior of Yersinia enterocolitica in Foods

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Yersinia enterocolitica are ubiquitous, being isolated frequently from soil, water, animals, and a variety of foods. They comprise a biochemically heterogeneous group that can survive and grow at refrigeration temperatures. The ability to propagate at refrigeration temperatures is of considerable significance in food hygiene. Virulent strains of *Yersinia* invade mammalian cells such as HeLa cells in tissue culture. Two chromosomal genes, *inv* and *ail*, were identified for cell invasion of mammalian. The pathogen can cause diarrhoea, appendicitis and post-infection arthritis may occur in a small proportion of cases. The most common transmission route of pathogenic *Y. enterocolitica* is thought to be fecal-oral via contaminated food. Direct person-to-person contact is rare. Occasionally, pathogenic *Y. enterocolitica* has been detected in vegetables and environmental water; thus, vegetables and untreated water are also potential sources of human yersiniosis. However, the isolation rates of pathogenic *Y. enterocolitica* have been low, which may be due to the limited sensitivity of the detection methods. To identify other possible transmission vehicles, different food items should be studied more extensively. Many factors related to the epidemiology of *Y. enterocolitica*, such as sources, transmission routes, and predominating genotypes remain obscure because of the low sensitivity of detection methods.